



Research Article

Enhanced dispersoids precipitation, recrystallization resistance and mechanical properties of Al-Mg-Si-Cu-Mn alloy via Mo addition

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ABSTRACT

The present study aims to clarify the influence of Mo addition on α -dispersoids precipitation, associated recrystallization resistance and mechanical performance of Mn-containing Al-Mg-Si-Cu alloys. Results reveal that joint addition of Mn and Mo as dispersoid-forming elements dramatically enhances α -dispersoids precipitation, achieving a fivefold increment in number density and a 26.5 % reduction in size compared to the alloy containing only Mn addition. Additionally, the width of dispersoid-free zones is reduced from appropriately 5 μm to <1 μm . A portion of the α -Al(Mn, Mo, Fe)Si dispersoids exhibits the quasicrystal characteristics, forming coherent interfaces with the aluminum matrix. In comparison, α -Al(Mn, Fe)Si dispersoids transformed entirely into a simple cubic structure, being partially coherent with the matrix. The improvements in α -dispersoids precipitation contribute to higher pinning force on moving boundaries, resulting in superior resistance to recrystallization of the alloy. Consequently, an ideal fibrous microstructure is retained in alloy co-alloyed with Mn and Mo after solution treatment. Whereas, alloys containing only Mn or Mo show complete recrystallization. A superior balance between strength and ductility is achieved for 0.7Mn0.3Mo alloy, attributed to the increased work hardening capacity and suppression of dynamic recovery.

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1. Introduction

Al-Mg-Si-Cu (6XXX) alloys are widely utilized in various industries due to their high strength-to-weight ratio, favorable formability, and outstanding corrosion resistance [1–4]. Precipitation strengthening is the primary mechanism for improving their mechanical strength, but this often comes at the expense of other properties, especially the ductility. The alloys are usually subjected to thermomechanical working to achieve the desired shape, during which, dynamic recovery results in the appearance of subgrain structures, which are highly desirable for their contribution to enhanced strength, toughness, and fatigue resistance [5,6]. Recrystallization, however, may occur during deformation or subsequent solution treatments, erasing these subgrain structures. Enhancing recrystallization resistance is therefore crucial for optimizing the overall performance of these alloys.

Nanoscale dispersoids precipitated during homogenization play a vital role in inhibiting recrystallization [2,7]. For instance, in

6XXX aluminum alloys, coherent $\text{Al}_3(\text{Sc}, \text{Zr})$ dispersoids are introduced to improve microstructure and performance [8,9]. Similarly, the addition of Mn promotes the formation of α -Al(Mn,Fe)Si dispersoids, which refine the deformation microstructure [10–13]. The retarding effect of dispersoids can be evaluated using the classical Zener pinning theory, which emphasizes the important role of dispersoids characteristics, such as the size and volume fraction [14]. Therefore, considerable researches have been conducted to improve dispersoids precipitation. Multi-step homogenization has proven effective in enhancing dispersoids precipitation and associated mechanical performance of aluminum alloys [15–17]. In addition, the precipitation characteristics of dispersoids are largely contingent upon the microalloying elements chosen. Additions of Cd and Sn were reported to enhance the number density of α -Al(Mn, Fe)Si dispersoids in Al-Mn-Si alloys by forming Cd- and Sn-rich nanoprecipitates, which provides favorable nucleation sites for the dispersoids [18,19]. The combined addition of dispersoid-forming elements can also influence precipitation behavior. While joint Mn and Zr additions in Al-Cu-Li alloys were found to reduce recrystallization resistance, despite the fact that both Mn and Zr can contribute to the formation of dispersoids [7,20]. This can be attributed to the increased Al_3Zr dispersoid-free band width and

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low effectiveness of $\text{Al}_{20}\text{Cu}_2\text{Mn}_3$ dispersoids in inhibiting recrystallization. In contrary, the synergistic addition of Mn and Cr significantly enhanced α -dispersoids precipitation due to complementary microsegregation patterns [10,21].

Mo, as a dispersoid-forming element, also contributes to improved mechanical properties in aluminum alloys. Studies have indicated the strengthening effect of Mo-bearing dispersoids in Al-Si-Cu-Mg [22,23] and Al-Mn-Mg alloys [24]. Farkoosh et al. [22,23] reported the formation of nano-sized α -Al(Mo, Fe)Si dispersoids in Al-Si-Cu-Mg-Mo alloy during solid solution treatment. In response to the excellent thermal stability, the Mo-containing alloy exhibited the improved creep resistance. Furthermore, Mo was found to enhance the number density and uniformity of α -dispersoids when combined with Mn. The enhanced elevated-temperature properties resulted from Mo addition was also reported in Al-Mn-Mg [24] and Al-Cu-Mg-Ag-Mn alloys [25]. It can be found that the addition of Mo not only enables the formation of α -Al(Mo, Fe)Si dispersoids, but also has a significant impact on the precipitation of α -Al(Mn, Fe)Si in Mn-containing alloys. Consequently, Mo addition also provides a possibility to improve the dispersoids precipitation in 6XXX aluminum alloys. However, conflicting results exist regarding that, with some studies suggesting a reduction in dispersoid formation due to decreased Mn and Si supersaturation [13]. This inconsistency underscores the need for further research on Mo microalloying in 6XXX alloys.

This study aims to elucidate the influence of Mo addition on the microstructure evolution and mechanical properties of Al-Mg-Si-Cu-Mn alloys. Comprehensive analyses were conducted on three alloys with individual and joint Mn and Mo additions, focusing on the precipitation characteristics of dispersoids, including their spatial distribution, number density, size et al. The resultant influence on deformed microstructure and mechanical performance was systematically investigated. The underlying influence mechanism was discussed based on Zener pinning theory and elaborate HRTEM analysis of the α -dispersoids.

2. Experimental procedure

Three Al-Mg-Si-Cu alloys with varying Mn and Mo additions were cast in a water-cooled copper mold (25 mm × 100 mm × 200 mm). The chemical compositions, listed in Table 1, were analyzed using a Spectro optical emission spectroscopy. Homogenization was performed in a Nabertherm N120/85HA air circulating resistance furnace at 560 °C for 10 h (with a heating rate of 3 °C/min), followed by water quenching. Microstructure analysis was conducted using a Jeol JSM-IT800 scanning electron microscope (SEM) and a Thermo Scientific Talos F200X transmission electron microscope (TEM). TEM foils were prepared using the twin-jet method in a methanol solution containing 25 % nitric acid, with a voltage of 10 V and temperatures ranging from −20 to −30 °C. TEM examinations were performed at 200 kV along the $\langle 001 \rangle_{\text{Al}}$ zone axis. Energy dispersive X-ray detector in HAADF-STEM mode was used to analyze dispersoids compositions, with a dwell time of 32 μs and a defocus of −3.8 μm . The Nano Measurer software was employed to quantify the number density and equivalent diameter of dispersoids across at least five TEM images. Dispersoid structures were characterized using high-resolution TEM (HRTEM), with corresponding fast Fourier

Table 1
Chemical compositions of the investigated alloys (wt%).

Alloys	Mg	Si	Cu	Mn	Mo	Fe	Ti	Al
0.7Mn	0.85	0.95	0.66	0.79	–	0.05	0.02	Bal.
0.3Mo	0.87	0.95	0.61	–	0.23	0.06	0.02	Bal.
0.7Mn0.3Mo	0.89	0.96	0.60	0.76	0.25	0.05	0.03	Bal.

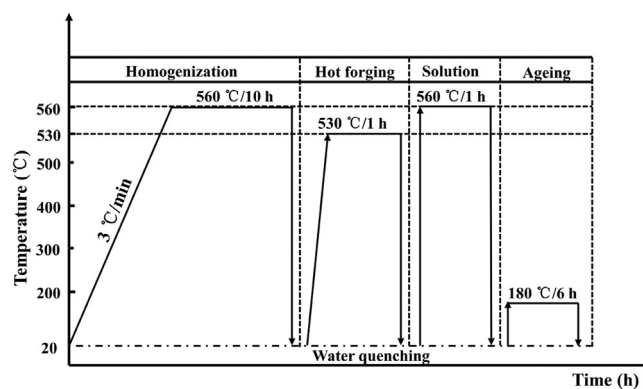


Fig. 1. Schematic diagram of the heat treatment and hot deformation processes.

transform (FFT) patterns acquired via Velox software. TEM foil thickness was determined using the CBED methods.

To evaluate the impact of Mo addition on deformed microstructure and mechanical properties, samples with a dimension of 20 mm × 20 mm × 160 mm were subjected to open-die forging at 530 °C with 70 % deformation, followed by water quenching and conventional T6 heat treatment (solution treatment at 560 °C for 1 h, aging at 180 °C for 6 h, as depicted in Fig. 1). It should be noted that the uneven deformation across the thickness of samples will result in uneven deformed microstructure. To avoid the resulted influence, the microstructural analysis and tensile tests are performed on the specimen taken at the center of the samples, as referred to previous work [17]. Tensile tests were performed at room temperature using a Shimadzu AGS-X universal tensile testing machine at a crosshead speed of 2 mm/min. Fracture morphology was examined using SEM. Deformed microstructure was analyzed using electron backscattered diffraction (EBSD) at 20 kV with a step size of 0.3 μm . OIM software was used for EBSD data analysis, with low-angle grain boundaries (LAGBs, 2°–15°) and high-angle grain boundaries (HAGBs, >15°) represented by white and black lines, respectively. Recrystallization fractions were quantified using Image-Pro Plus software.

3. Results

3.1. As-cast microstructure

Fig. 2 illustrates the backscattered SEM images of the as-cast microstructure of the investigated alloys, showing no significant difference across the samples. The microstructure consists of α -Al and interdendritic intermetallic compounds. EDX mapping confirms that the dark intermetallic compounds are Mg_2Si , the interdendritic spherical bright phases are Q-AlMgSiCu, while the bright platelet-like phases distributed along grain and dendritic cell boundaries are characterized as Fe-bearing intermetallics. The composition of the Fe-bearing phases varies with Mn and Mo additions, forming Al(Mn, Fe)Si, Al(Mo, Fe)Si, and Al(Mn, Mo, Fe)Si phases, respectively. Following homogenization treatment, all Mg_2Si and Q-AlMgSiCu phases dissolve back into the aluminum matrix, leaving only a portion of the high-melting-point Fe-bearing phases, as shown in Fig. 3. Mo, as a peritectic element, can form large primary AlMo phase during solidification when added in excess, potentially degrading mechanical properties [22,23]. In this study, no primary AlMo phases were detected, suggesting that the Mo content was appropriately chosen.

3.2. Dispersoids

SEM micrographs in Fig. 4 reveal the large numbers of dispersoids precipitated in the investigated alloys after homogenization.

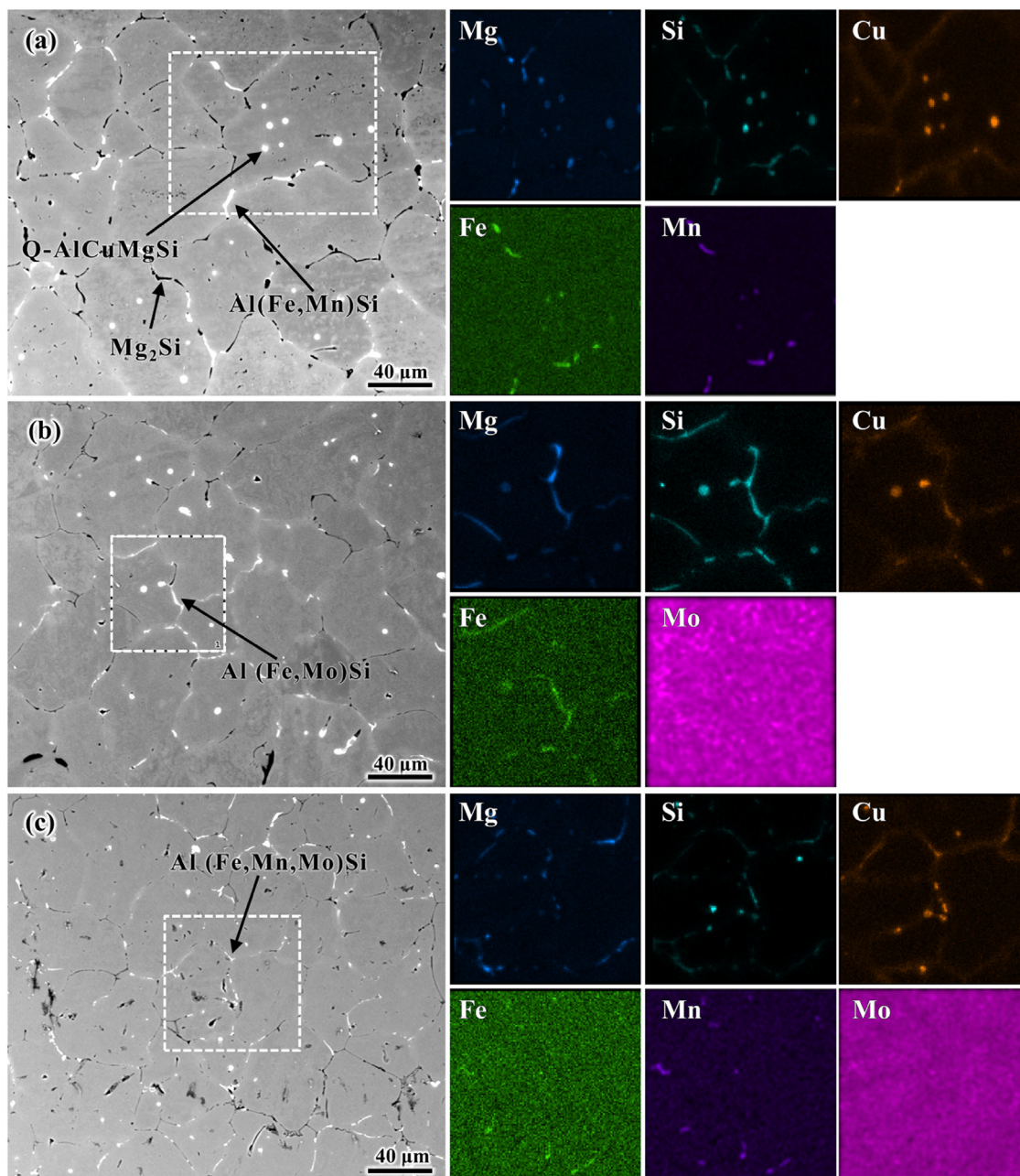


Fig. 2. Backscattered SEM- images and EDX mapping of the intermetallics formed in the as-cast (a) 0.7Mn, (b) 0.3Mo, and (c) 0.7Mn0.3Mo alloys.

The 0.3Mo alloy exhibits sparse dispersoid precipitation primarily near grain and dendritic cell boundaries. In contrast, the 0.7Mn and 0.7Mn0.3Mo alloys display more uniform dispersoid distributions with higher densities, with the 0.7Mn0.3Mo alloy exhibiting the densest precipitation. These results demonstrate that the co-addition of Mn and Mo significantly improves dispersoid precipitation in Al-Mg-Si-Cu alloys.

To determine the compositions of dispersoids precipitated with various Mn and Mo additions, a detailed examination was conducted using STEM-EDX, and the results are illustrated in Fig. 5. A considerable number of nano-sized dispersoids with various morphologies can be clearly observed in the HAADF STEM images. The EDX elemental maps indicate that when only Mn is added, the dispersoids are composed of Mn, Fe, and Si. The dispersoids in 0.3Mo alloy consist of Mo, Fe, and Si. While in the 0.7Mn0.3Mo alloy, the dispersoids are enriched with Mn, Mo, Fe, and Si. Based on these findings, the dispersoids precipitated in the 0.7Mn, 0.3Mo, and

0.7Mn0.3Mo alloys are suggested to be α -Al(Mn, Fe)Si, α -Al(Mo, Fe)Si, and α -Al(Mn, Mo, Fe)Si, respectively.

Bright-field TEM images in Fig. 6, along with size distribution data, reveal substantial differences in dispersoid precipitation across the alloys. Individual addition of 0.3Mo exhibits a negligible number of dispersoids with extremely uneven distribution. In contrast, both the 0.7Mn and 0.7Mn0.3Mo alloys show substantial dispersoids precipitation with uniform distribution. Notably, the joint Mn and Mo additions significantly enhances the quantity of dispersoids while also notably reduces their size. The size of α -Al(Mn, Fe)Si dispersoids concentrates in the range of 100–200 nm, and the average equivalent diameter is determined to be 155 nm, as illustrated in Fig. 7. The Mo addition significantly reduce the dispersoids size, with the Mo-bearing dispersoids exhibit a predominance of approximately 100 nm (Fig. 6(e and f)), and the average size of α -Al(Mn, Mo, Fe)Si dispersoids is reduced to be 114 nm. This suggests that the introduction of Mo into Mn-containing al-

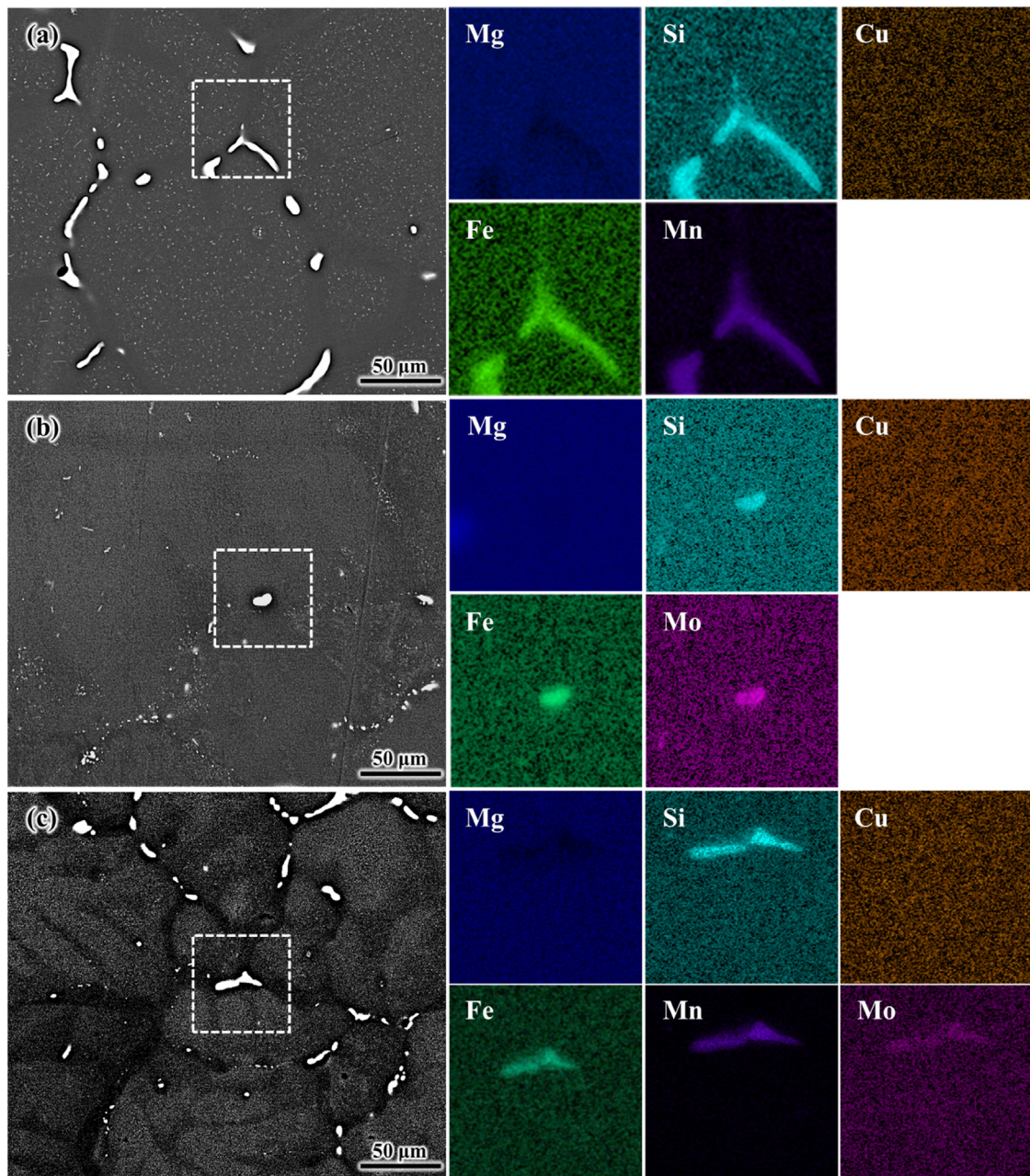


Fig. 3. Backscattered SEM images and EDX mapping of the intermetallics in (a) 0.7Mn, (b) 0.3Mo, and (c) 0.7Mn0.3Mo alloys after homogenization.

loy retards the Oswald ripening of the α -dispersoids due to its relatively lower diffusion coefficient in aluminum. However, despite the smallest size of α -Al(Mo, Fe)Si dispersoids in 0.3Mo alloy, the number density is determined to be only $5.43 \mu\text{m}^{-3}$. In comparison, the 0.7Mn alloy has a relatively higher number density of $8.45 \mu\text{m}^{-3}$. Notably, the co-addition of Mn and Mo significantly increases the number density of dispersoids, reaching $45.54 \mu\text{m}^{-3}$. These results are consistent with the findings of Farkoosh et al. in Al-Si-Cu-Mg alloy [23], which was ascribed to the opposite microsegregation patterns of Mn and Mo elements.

3.3. Deformed microstructure

To investigate the impact on deformed microstructure resulting from the observed dispersoids variation, the inverse pole figure (IPF) maps and corresponding misorientation angle distribution after deformation are presented in Fig. 8. All alloys exhibit the char-

acteristic of “fibrous” microstructure, characterized by elongated grains perpendicular to the forging direction and a high fraction of LAGBs (Fig. 8(d)). However, localized recrystallization is observed in the 0.7Mn alloy, which can be clearly identified in Fig. 8(a). The dynamic recrystallization is more severe in 0.3Mo alloy, displaying significant microstructure inhomogeneity accompanied with some large recrystallized grains. This can be attributed to the uneven precipitation of α -Al(Mo, Fe)Si dispersoids shown in Fig. 6(b). Regions with dispersoid precipitation retain fibrous microstructures, whereas regions lacking dispersoids experience severe dynamic recrystallization. In contrast, the 0.7Mn0.3Mo alloy achieves an ideal deformed microstructure with minimal dynamic recrystallization, highlighting the synergistic effect of Mn and Mo in enhancing recrystallization resistance.

Following solution treatment, as shown in Fig. 9, the differences in microstructure become more pronounced. Both the 0.7Mn and 0.3Mo alloys exhibit nearly complete recrystallization accompanied

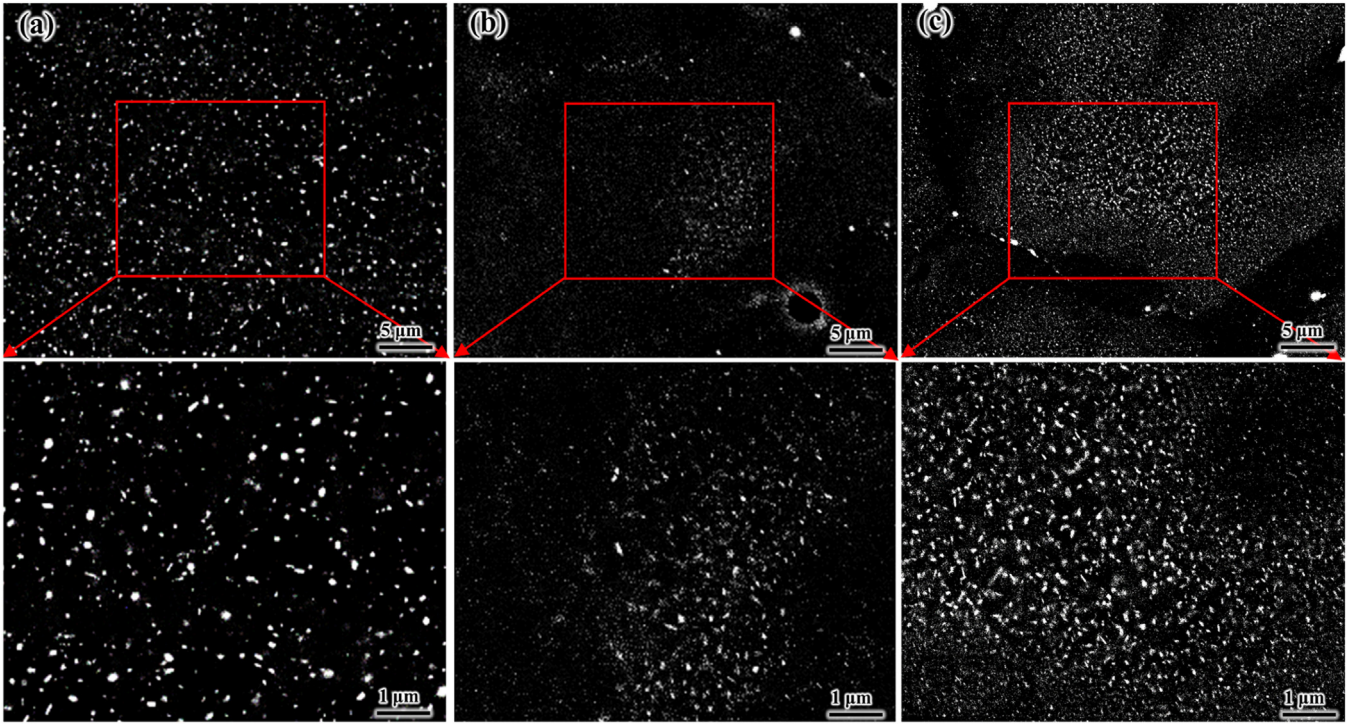


Fig. 4. SEM micrographs of the investigated alloys after homogenization: (a) 0.7Mn, (b) 0.3Mo, (c) 0.7Mn0.3Mo.

by significant grain growth, with the recrystallization fractions significantly increased to 0.81 and 0.85, respectively. This suggests that dispersoids in these alloys are insufficient to effectively inhibit the migration of dislocations and subgrain boundaries. In contrast, the 0.7Mn0.3Mo alloy maintains its fibrous microstructure with only minor changes in the fraction of LAGBs and recrystallization, further demonstrating the superior recrystallization resistance provided by the combined Mn and Mo additions.

3.4. Mechanical properties

The mechanical properties of the alloys, illustrated in Fig. 10, correlate strongly with their microstructures. The 0.3Mo alloy exhibits a yield strength of 372 MPa but an extremely low elongation of 1.9 %. In comparison, the 0.7Mn alloy exhibits improved mechanical properties with a yield strength of 387 MPa and an elongation of 7.5 %. And the properties are further improved with the joint Mn and Mo additions, resulting in the best strength-ductility balance in the 0.7Mn0.3Mo alloy.

4. Discussion

4.1. Relationship between microstructure and mechanical property

Precipitation hardening is known to be the primary strengthening mechanism for the Al-Mg-Si-Cu alloys. However, the three investigated alloys contain the same contents of Mg, Si, and Cu, which serve as the forming elements of the nanoscale precipitates during artificial aging. Consequently, the contributions of precipitation hardening for the three alloys should be identical, and the detected difference in the mechanical properties arises from the variation in the dispersoids and deformed microstructure. The dispersoids can contribute to strength enhancement by acting obstacle to dislocation motion, and this can be estimated using the Orowan bypassing mechanism [26]:

$$\Delta\sigma_{\text{Orowan}} = \frac{0.84MGb}{2\pi(1-\nu)^{0.5}\lambda} \ln\left(\frac{r}{b}\right) \quad (1)$$

$$\lambda = r \left(\frac{2\pi}{3f} \right)^{0.5} \quad (2)$$

where M (taken as 3.06) is the Taylor factor, G (27.4 GPa) is the shear modulus of the Al matrix, b (0.286 nm) is the Burgers vector, ν (0.34) is Poisson's ratio, λ is the effective interparticle spacing, r is the average equivalent radius of the dispersoids, and f is the volume fraction of dispersoids. The volume fraction f of dispersoids can be determined as following:

$$f = A_A \frac{KD}{KD+t} (1 - A_{\text{DFZ}}) \quad (3)$$

where A_A is the area fraction of dispersoids observed in TEM images, K is the average shape factor of dispersoids, D denotes the average equivalent diameter of dispersoids, A_{DFZ} denotes the area fraction of DFZ, and t corresponds to the thickness of TEM foils.

Based on the strengthening model, the strength contributions resulted from dispersoids in 0.7Mn and 0.3Mo alloys are calculated to be 14.39 and 10.36 MPa, respectively. While this contribution increases to 24.87 MPa for the 0.7Mn0.3Mo alloy due to the enhanced precipitation of dispersoids.

The mechanical performance of aluminum alloys is strongly influenced by their fracture modes. To clarify the relationship between deformed microstructure and mechanical properties, the fracture morphology of the alloys was analyzed. As shown in Fig. 11(b), the dominant fracture mode in 0.3Mo alloy is intergranular brittle fracture, a mode that significantly compromises its mechanical performance, especially the ductility. This can be attributed to the large recrystallized grains and the sparse precipitation of dispersoids, which provide minimal hindrance to dislocation motion. Consequently, dislocations accumulate rapidly at grain boundaries, leading to the initiation and propagation of intergranular cracks. The fracture performance of 0.7Mn alloy exhibits a mixture mode of brittle and ductile, with localized regions of brittle fracture clearly identified. Nevertheless, the mechanical properties of the 0.7Mn alloy are noticeably improved compared to the 0.3Mo alloy, as evidenced by its enhanced overall strength.

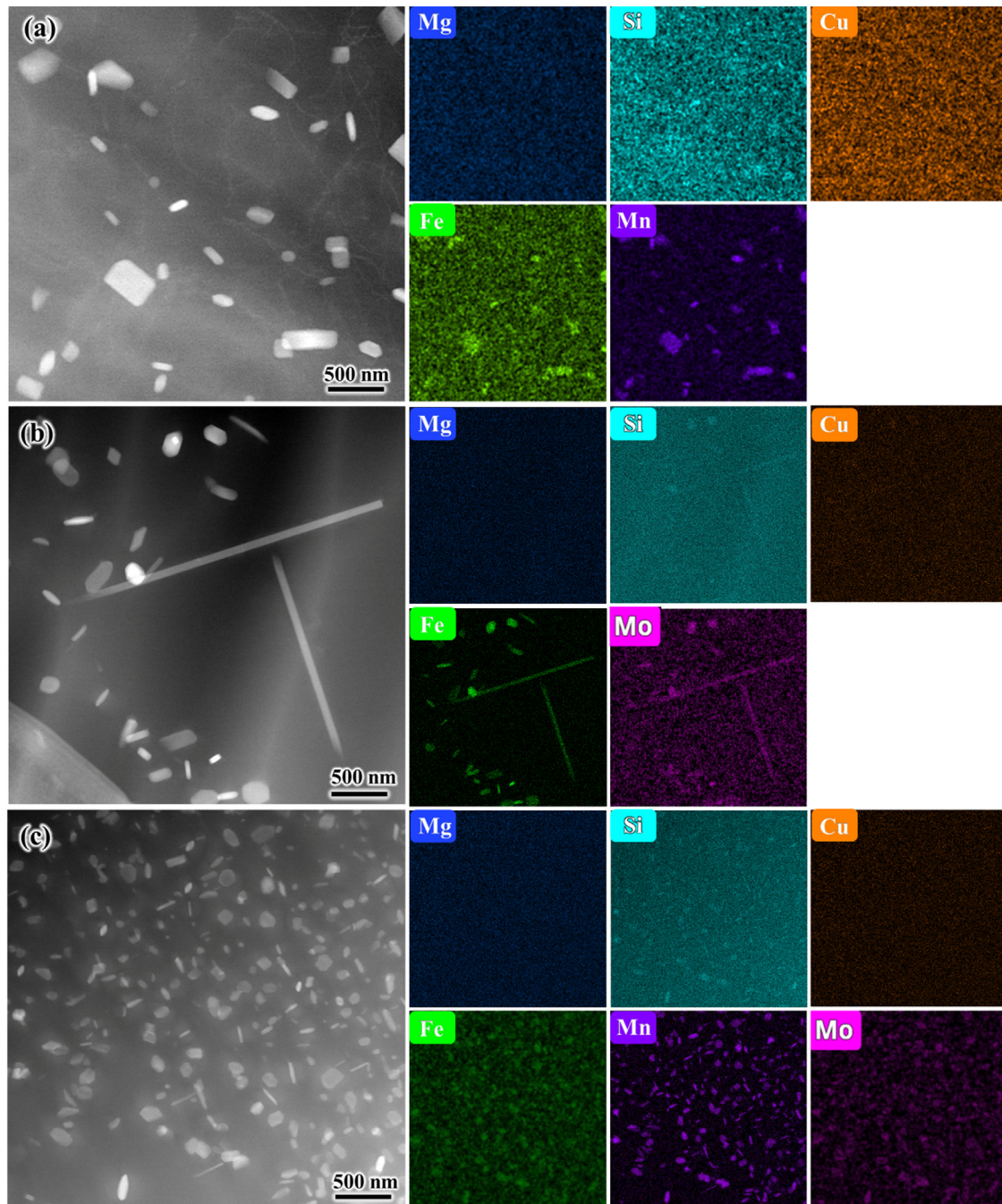


Fig. 5. HAADF STEM images and STEM-EDX elemental maps of dispersoids in (a) 0.7Mn, (b) 0.3Mo, and (c) 0.7Mn0.3Mo alloys.

In contrast, the 0.7Mn0.3Mo alloy, benefiting from a favorable fibrous microstructure, predominantly exhibits transgranular ductile fracture. This is characterized by the presence of dense and fine dimples (Fig. 11(c)), which absorb significant energy during tensile test. Consequently, the 0.7Mn0.3Mo alloy achieves an improved balance between strength and ductility, highlighting the synergistic effect of the combined Mn and Mo additions in enhancing the alloy's mechanical performance.

The work hardening behavior during plastic deformation, closely relating to microstructure characteristics, plays a significant role in the mechanical performance of alloys [27–29]. To investigate this, the true stress–strain curves and work hardening rate ($\theta = d\sigma/d\varepsilon$, σ and ε represent the true stress and true strain,

respectively) as functions of true strain for the investigated alloys are presented in Fig. 12(a). Meanwhile, the Kocks–Mecking plots illustrating the relationship between instantaneous work hardening rate and stress increment ($\sigma - \sigma_y$, σ_y is the yield strength) are shown in Fig. 12(b). The sharp drop in θ before reaching an inflection point is associated with the elastic-to-plastic transition. Following this transition, θ gradually decreases as flow stress increases until failure. In the case of the 0.3Mo alloy, θ rapidly decreases even during plastic deformation, with a stress increment of only 17 MPa before necking, indicating a very low work hardening capacity. As discussed earlier, the geometrically necessary dislocations generated during the initial plastic deformation stage rapidly move and accumulate at grain boundaries due to the lack of ob-

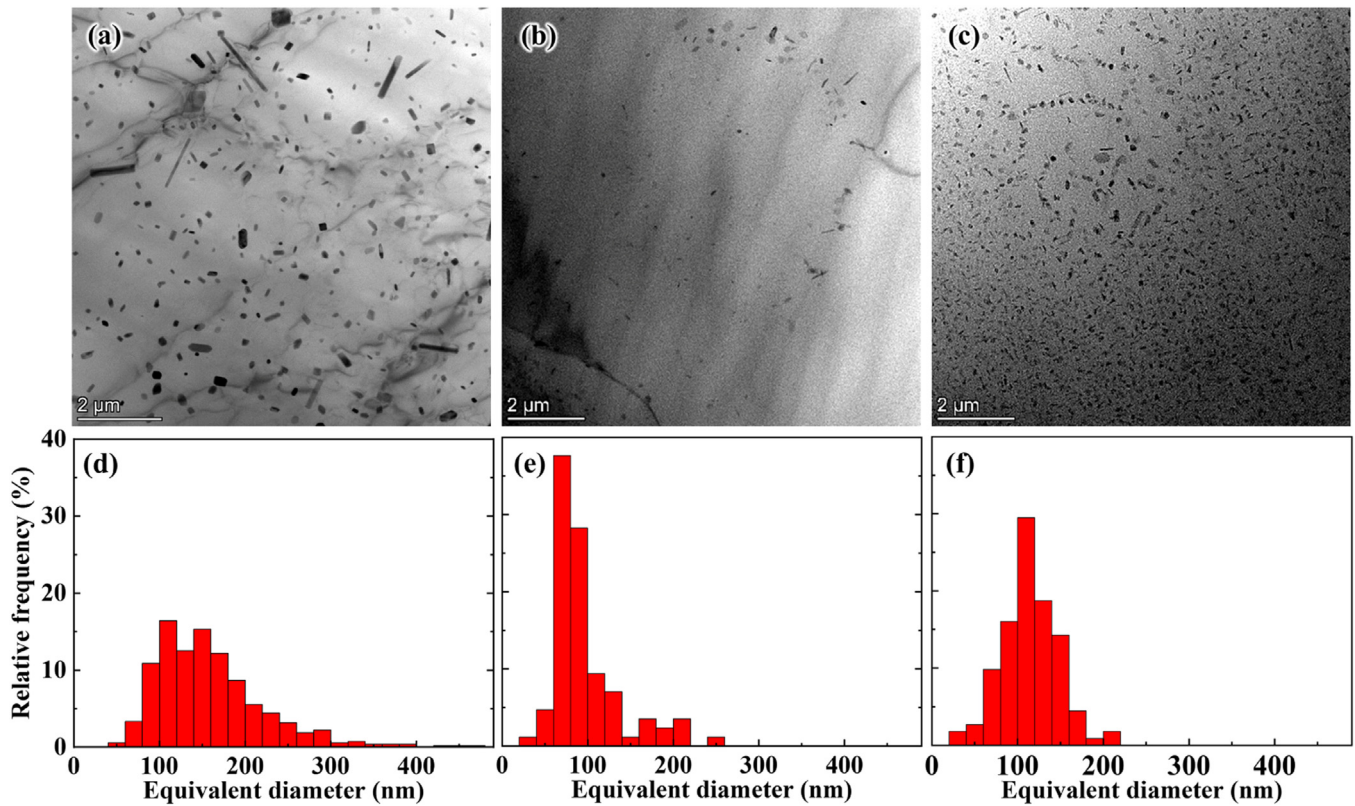


Fig. 6. Bright-field TEM images showing the dispersoids viewed along $[001]_{Al}$ direction and corresponding statistic size distribution in the alloys: (a, d) 0.7Mn, (b, e) 0.3Mo, and (c, f) 0.7Mn0.3Mo.

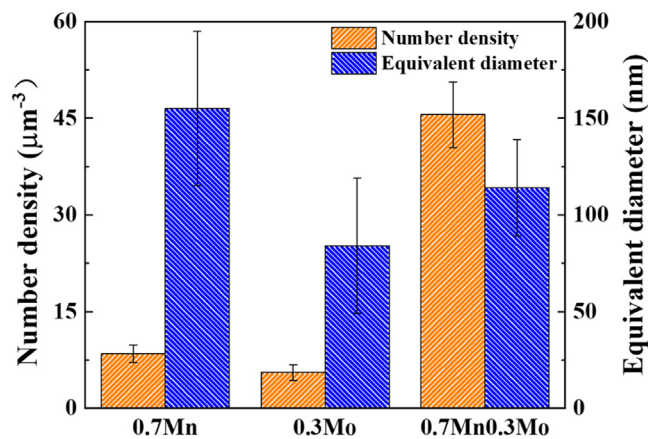


Fig. 7. Quantitative statistics of number density and equivalent diameter of the dispersoids.

stacles, leading to stress concentration and subsequent early failure. In contrast, enhanced work hardening capacity is achieved for 0.7Mn alloy, with an increased stress increment of 46.8 MPa despite the occurrence of severe recrystallization. This improvement can be attributed to the higher quantity of dispersoids in 0.7Mn alloy, where dispersoid–dislocation interactions lower stress concentration at grain boundaries, thereby mitigating the risk of intergranular fracture. This is consistent with the findings of Arani et al. that complete intergranular fracture occurs in dispersoid-free alloys, whereas alloys containing dispersoids exhibit predominantly transgranular fracture [30]. A superior work hardening capacity is achieved for 0.7Mn0.3Mo alloy, as evidenced by a flatter Kocks–Mecking plot and a stress increment of 51.5 MPa be-

fore necking. Moreover, the parameter β ($-d\theta/d\sigma$), derived from the Kocks–Mecking plots, is used to characterize dynamic recovery [31], which occurrence hinders dislocation accumulation during plastic deformation. The β value for the 0.7Mn0.3Mo alloy is significantly lower than those of the other two alloys, suggesting the suppressed dynamic recovery. This is ascribed to the further enhanced precipitation of α -Al(Mn, Mo, Fe)Si dispersoids and the introduction of subgrains, both of which hinder the formation of dislocation cells and retain a high work hardening rate under high stress during deformation [27]. Therefore, due to the enhanced work hardening capacity and delayed plasticity instability, the alloy with co-addition of Mn and Mo achieves a superior balance between strength and ductility.

Schmid factor is commonly employed to explore the correlations between deformed microstructures and associated fracture modes, as it delineates the minimum shear stress required to activate a slip system in crystalline materials. The $\{111\} \langle 1-10 \rangle$ Schmid factor distribution maps corresponding to the deformed microstructure after solution treatment (Fig. 9) are shown in Fig. 13 for the investigated alloys. The results clearly present relatively higher Schmid factor values for the 0.7Mn and 0.3Mo alloys, particularly in regions with abnormally large recrystallized grains, as denoted by the arrows. This suggests that the large recrystallized grains are relatively “softer”, as a high Schmid factor correlates with a greater likelihood of slip system activation during tensile tests [32]. Consequently, the reduced resistance to plastic deformation makes the 0.3Mo and 0.7Mn alloys more susceptible to deformation, with the subsequent pile-up of dislocations at high-angle grain boundaries [33–35], leading to the nucleation of intergranular cracks. In contrast, benefiting from the well-maintained fibrous microstructure consisting of numerous subgrains, the 0.7Mn0.3Mo alloy experiences more uniform deformation, which contributes to its improved ductility.

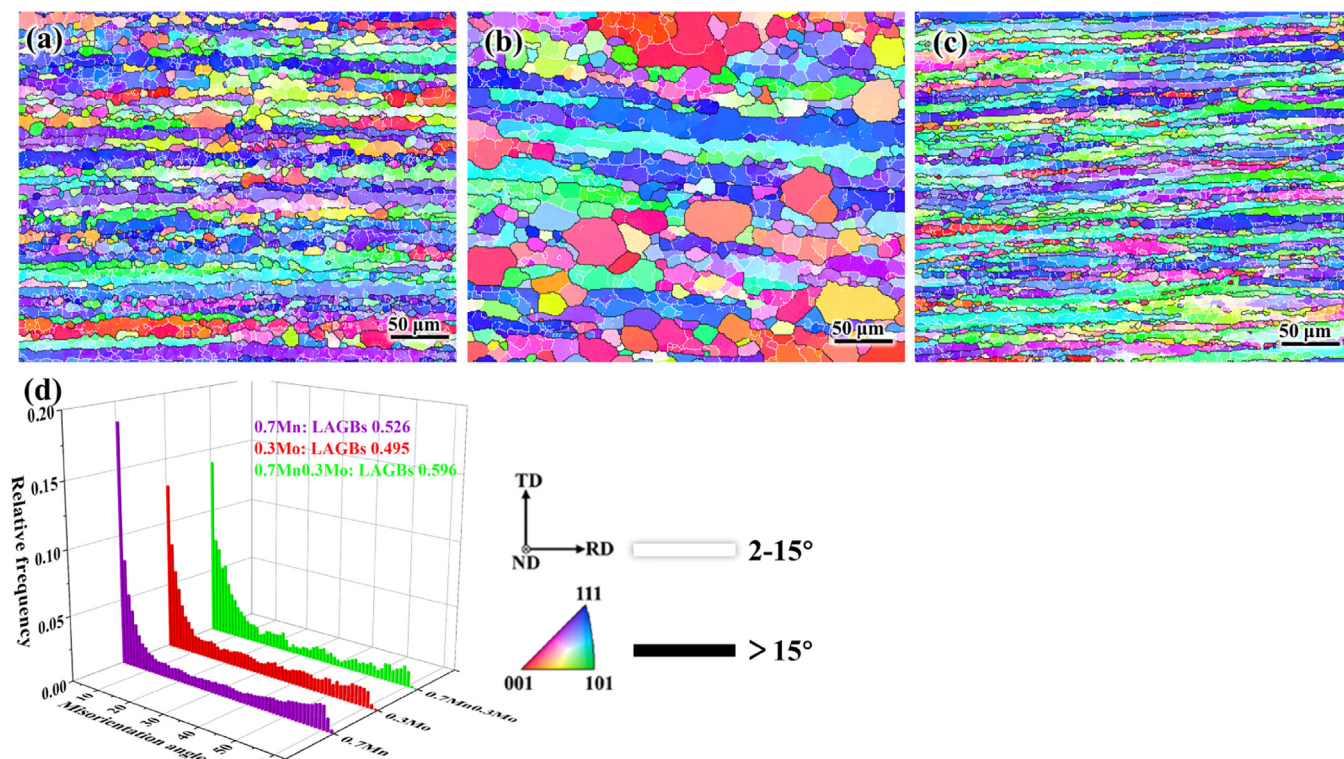


Fig. 8. EBSD IPF maps of (a) 0.7Mn, (b) 0.3Mo, (c) 0.7Mn0.3Mo alloys after deformation without solution treatment. (d) Misorientation angle distribution.

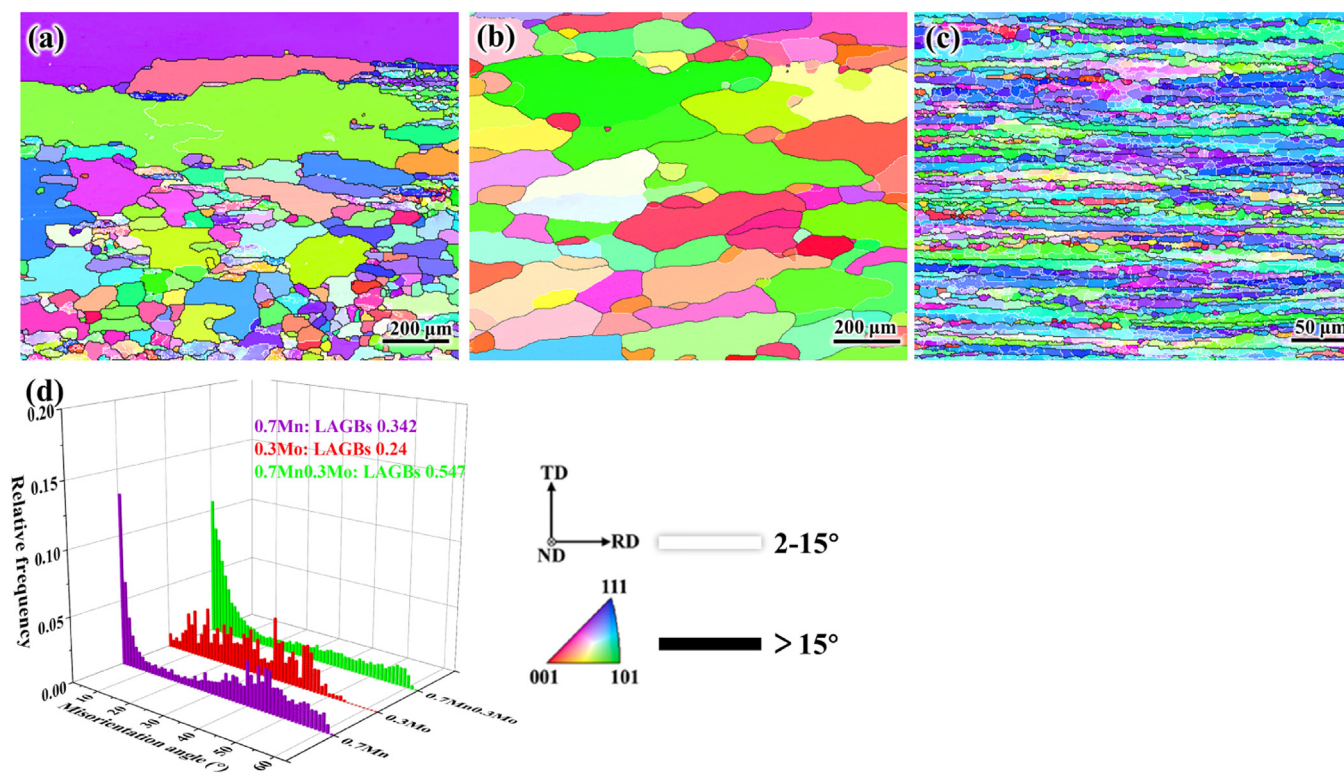


Fig. 9. EBSD IPF maps of (a) 0.7Mn, (b) 0.3Mo, (c) 0.7Mn0.3Mo alloys after solution treatment. (d) Misorientation angle distribution.

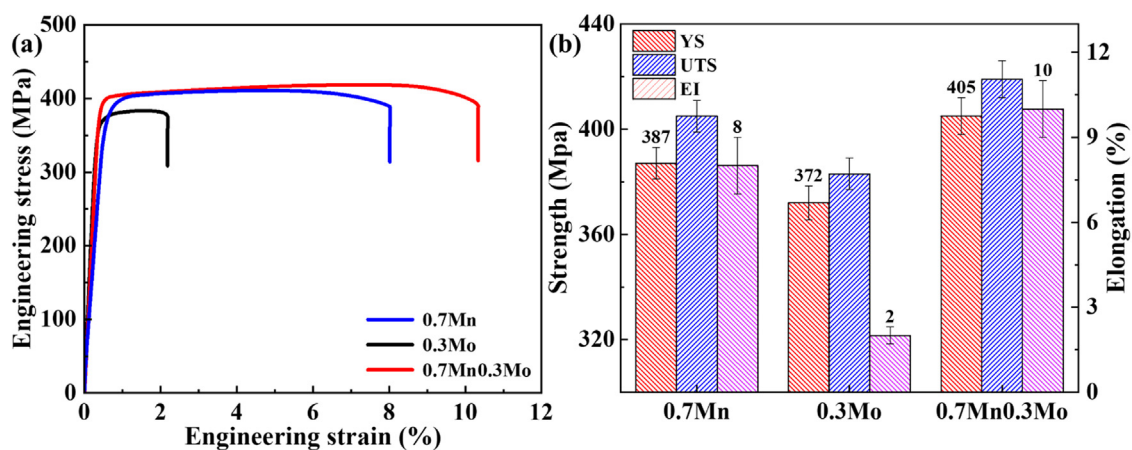


Fig. 10. Mechanical properties of the experimental alloys after T6 treatment: (a) engineering stress–strain curves, (b) corresponding data.

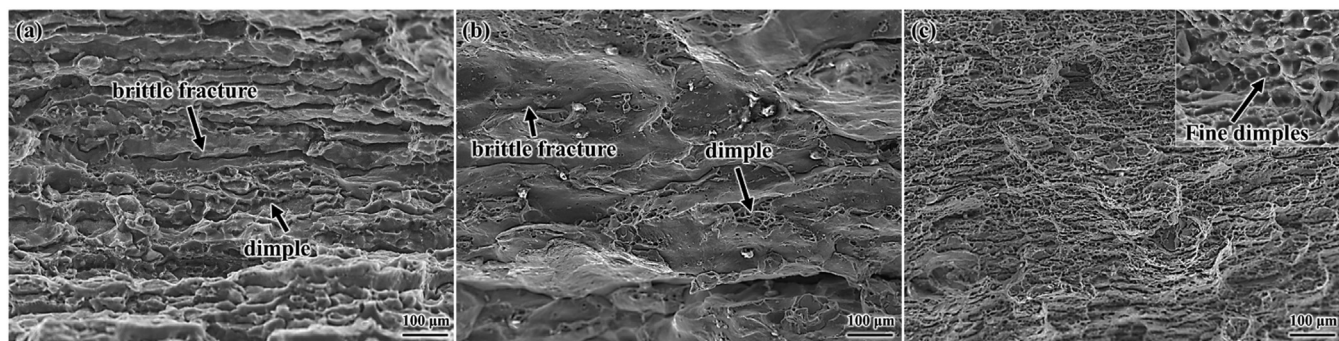


Fig. 11. SEM micrographs showing the fracture surfaces of (a) 0.7Mn, (b) 0.3Mo, and (c) 0.7Mn0.3Mo alloys.

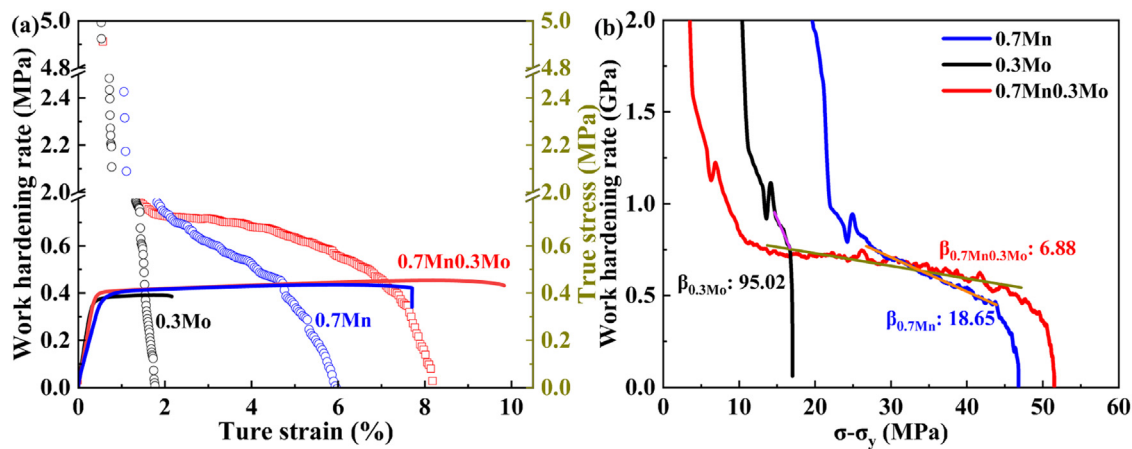


Fig. 12. Plots of work hardening rate and true stress as a function of (a) true strain, (b) stress increments for the alloys.

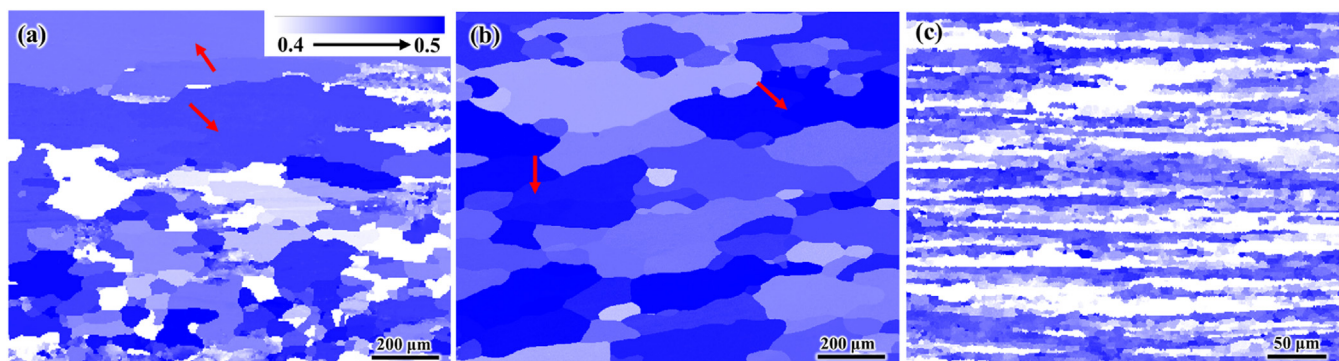


Fig. 13. {111} <1-10> Schmid factor distribution maps of (a) 0.7Mn, (b) 0.3Mo, and (c) 0.7Mn0.3Mo alloys.

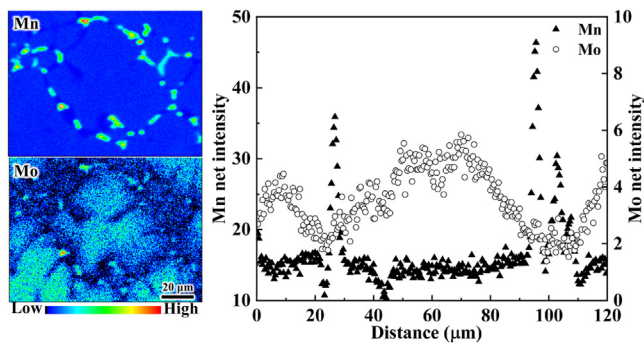


Fig. 14. Composition maps and profiles of Mn and Mo elements within an as-cast grain.

4.2. Dispersoids precipitation

The joint addition of Mn and Mo remarkably enhances the precipitation characteristics of α -dispersoids compared to alloys with sole Mn or Mo addition. Similar improvements in dispersoid precipitation have been reported with the co-addition of Mn and Cr, where the enhancement was attributed to the opposing microsegregation behaviors of Mn and Mo/Cr during solidification [22]. This specific microsegregation feature of Mn and Mo can be clearly identified in Fig. 14, which shows the distribution of these elements within an as-cast grain, as obtained by EPMA. Mo segregates towards the grain interior, whereas Mn concentrates at the grain boundaries. In addition to Mo and Cr, a similar opposite microsegregation pattern also exists between Mn and Zr. However, joint additions of Mn and Zr in Al-Cu-Li alloys have been reported to yield relatively poorer recrystallization resistance compared to alloys containing only Zr [7,20]. This effect was attributed to the widened DFZs and decreased number density of Al_3Zr dispersoids in the presence of Mn [36]. While the reported opposing microsegregation patterns of Mn and Mo contribute to enhanced dispersoid precipitation, they may not be the sole factor. A more significant influence likely stems from the exceptionally low diffusion coefficient of Mo in the Al matrix. The diffusion coefficients of Mn and Mo at 400 °C were measured to be 6.24×10^{-19} and $5.52 \times 10^{-23} \text{ m}^2 \text{ s}^{-1}$, respectively [37]. The coefficient of Mo is four orders of magnitude lower than that of Mn, suggesting the prior aggregation tendency of Mn during homogenization. Consequently, $\alpha\text{-Al}(\text{Mn, Fe})\text{Si}$ dispersoids likely nucleate first in the Al matrix. Subsequently, Mo gradually diffuses into these dispersoids, leading to the formation of $\alpha\text{-Al}(\text{Mn, Mo, Fe})\text{Si}$. The introduction of Mo may also enhance the thermal stability of the dispersoids by suppressing the Oswald ripening process. This is directly related to the variation in dispersoid structure and dispersoid/matrix interfacial energy [38,39]. Although further investigations by APT and first principles calculations are necessary to confirm this in future studies, the retention of intermediate quasicrystals in the 0.7Mn0.3Mo alloy, which will be presented later, can serve as indirect evidence supporting this hypothesis.

4.3. Recrystallization resistance

A considerable quantity of dispersoids precipitated in the aluminum matrix after homogenization treatment, which plays a crucial role in retarding recrystallization through the pinning effect on dislocations and grain boundaries. The effectiveness of this pinning effect can be quantitatively evaluated using the Zener drag theory [14]:

$$P_z = 3\gamma f/D \quad (4)$$

where P_z represents the Zener pinning force, γ represents the grain boundary energy.

Table 2

Average size, volume fraction, and associated pinning force resulted from the dispersoids in the investigated alloys.

Alloys	Average size, D (nm)	Volume fraction, f (%)	Pinning force, P_z
0.7Mn	155	0.563	$1.09 \times 10^{-4} \gamma$
0.3Mo	84	0.108	$0.39 \times 10^{-4} \gamma$
0.7Mn0.3Mo	114	1.018	$2.68 \times 10^{-4} \gamma$

According to the Zener drag theory, the f/D value primarily determines the pinning effect of the dispersoids. Therefore, a higher f/D value is essential to enhance the recrystallization resistance of the alloys. Table 2 presents the average size, volume fraction, and corresponding pinning forces for the alloys under investigation. The 0.3Mo alloy exhibits the smallest pinning force, primarily due to the very low volume fraction of the $\alpha\text{-Al}(\text{Mo, Fe})\text{Si}$ dispersoids, leading to the poorest recrystallization resistance. In comparison, the 0.7Mn alloy shows a relatively higher pinning force, while the 0.7Mn0.3Mo alloy achieves the highest pinning force due to the favorable precipitation characteristics of $\alpha\text{-Al}(\text{Mn, Mo, Fe})\text{Si}$ dispersoids, including a high volume fraction and small size. In response to that, the alloy with co-addition of Mn and Mo possesses superior recrystallization resistance.

Despite the high pinning force resulting from the large numbers of dispersoids, the spatial distribution of these dispersoids also plays a significant role in the pinning effect. The dispersoids are primarily concentrated within the interior of the grains, leading to the formation of dispersoid-free zones (DFZs) near grain boundaries. These DFZs, characterized by an extremely low pinning force, serve as preferential sites for recrystallization. Therefore, eliminating or reducing width of the DFZs is beneficial for enhancing the recrystallization resistance of alloys [40,41]. The HAADF-STEM images in Fig. 15 illustrate the characteristic of DFZs in alloys with varying Mn and Mo additions. The 0.7Mn alloy exhibits DFZs with a width greater than 5 μm , while the width of the DFZs in 0.7Mn0.3Mo alloy is significantly reduced to below 1 μm . As a consequence, the favorable DFZs characteristics in 0.7Mn0.3Mo alloy contribute significantly to its enhanced recrystallization resistance.

In addition to the previously discussed characteristics of the dispersoids, the interface characteristics between the dispersoids and the Al matrix also markedly influence their pinning effects on moving boundaries [14,42]. To investigate this, typical HRTEM micrographs and corresponding FFT patterns of the various dispersoids are illustrated in Fig. 16. As shown in Fig. 16(a), the $\alpha\text{-Al}(\text{Mn, Fe})\text{Si}$ dispersoids show a simple cubic structure, with an orientation relationship with Al matrix being $[100]_\alpha // [100]_{\text{Al}}$, $(054)_\alpha // (002)_{\text{Al}}$, suggesting a partially coherent interface with the Al matrix. Similarly, the $\alpha\text{-Al}(\text{Mo, Fe})\text{Si}$ and $\alpha\text{-Al}(\text{Mn, Mo, Fe})\text{Si}$ dispersoids also share the simple cubic structure and partially coherent relationship with Al matrix. But interestingly, some quasicrystal $\alpha\text{-Al}(\text{Mn, Mo, Fe})\text{Si}$ dispersoids are detected in the 0.7Mn0.3Mo alloy, as shown in Fig. 16(d). These dispersoids exhibit an icosahedral quasicrystal structure, previously reported in the literature [19], and demonstrate a coherent interfacial relation with the Al matrix. The coherent dispersoids are believed to contribute a stronger pinning force on moving boundaries due to their relatively lower interfacial energy [14]. Quantitatively, the maximum pinning force associated with a coherent dispersoid is twice that of an incoherent dispersoid of the same size [21]. Consequently, the presence of coherent quasicrystal $\alpha\text{-Al}(\text{Mn, Mo, Fe})\text{Si}$ dispersoids further improves the recrystallization resistance of 0.7Mn0.3Mo alloy. According to the studies by Hansen et al. [43,44], quasicrystals are recognized as the intermediate stage of Mn-bearing dispersoids during the precipitation process. Therefore, the existence of coherent quasicrystal $\alpha\text{-Al}(\text{Mn, Mo, Fe})\text{Si}$ dispersoids may be attributed to a de-

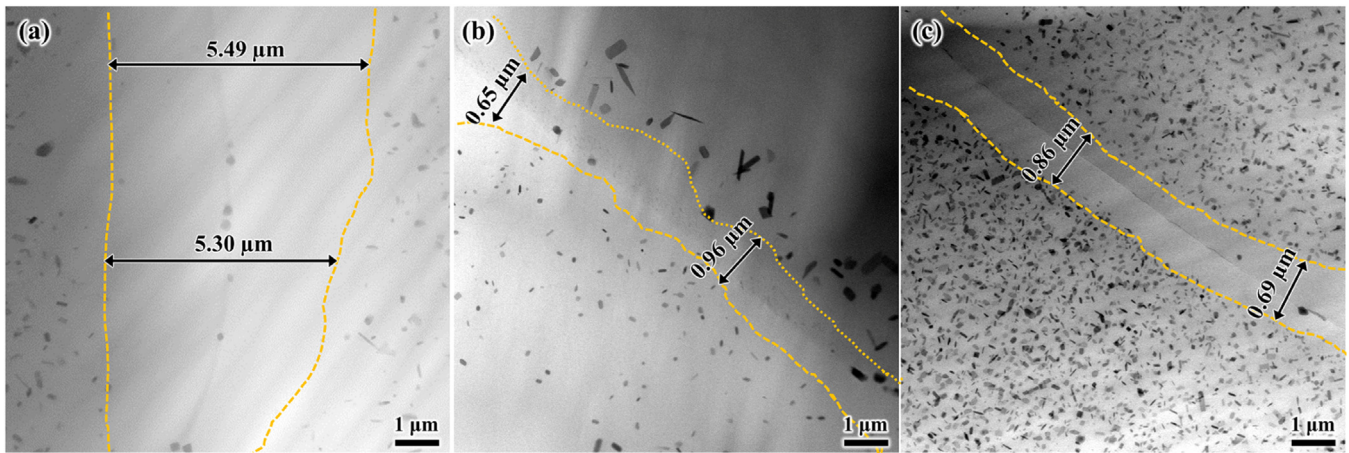


Fig. 15. HAADF-STEM micrographs showing the characteristic of dispersoid-free band in alloys (a) 0.7Mn, (b) 0.3Mo, and (c) 0.7Mn0.3Mo.

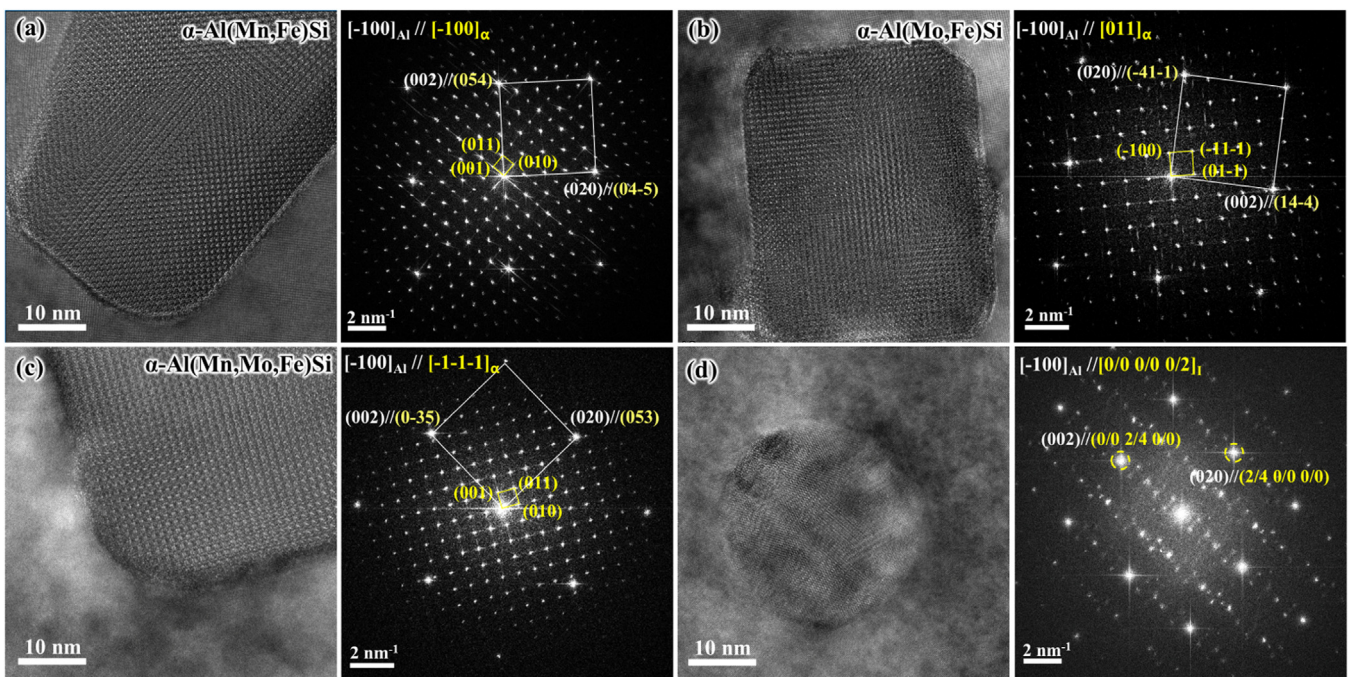


Fig. 16. HRTEM micrographs and corresponding FFT patterns of (a) α -Al(Mn,Fe)Si, (b) α -Al(Mo,Fe)Si, (c, d) α -Al(Mn,Mo,Fe)Si dispersoids.

layed transformation from intermediate quasicrystals to the final cubic phases. This hypothesis warrants further systematic investigation to elucidate the underlying mechanisms in future studies.

5. Conclusions

This study systematically investigates the effects of Mo addition on the precipitation of α -dispersoids in a Mn-containing Al-Mg-Si-Cu alloy, along with the associated recrystallization resistance and mechanical performance. The following conclusions are drawn:

- (1) The joint addition of Mn and Mo significantly enhances the precipitation of α -dispersoids compared to the alloy with sole addition. Specifically, the number density of dispersoids increases dramatically from $8.45 \mu\text{m}^{-3}$ to $45.54 \mu\text{m}^{-3}$, while the average dispersoid size decreases from 155 nm to 114 nm. The DFZs width is reduced from appropriately 5 μm to below 1 μm . Furthermore, a portion of coherent quasicrystal α -Al(Mn, Mo, Fe)Si dispersoids remains in the 0.7Mn0.3Mo alloy after homogenization. These improve-

ments collectively contribute to an increased pinning force on moving dislocations and grain boundaries.

- (2) Due to the enhanced pinning force, the alloy with the co-addition of Mn and Mo exhibits superior resistance to recrystallization, maintaining a well-preserved fibrous microstructure after solution treatment. In contrast, alloys with individual Mn or Mo addition undergo complete recrystallization under the same thermo-mechanical conditions.
- (3) The improved precipitation of α -dispersoids and the resultant maintenance of the fibrous microstructure enhance the work hardening capacity and suppress the dynamic recovery of the alloy during tensile deformation. Consequently, the 0.7Mn0.3Mo alloy achieves an enhanced balance between strength and ductility.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Xiang Li: Writing – original draft, Investigation, Data curation.
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Shihu Hu: Investigation, Data curation.
Xiaonan Wang: Investigation, Supervision.
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